
paper_id: PAPER_964
title: “3D MUGE Magnetar Simulation”
session: 215
date: 2026-04-12
author: “Daniel T. Murphy”
status: production
cvw: “v2.0.0”
tags: [AGN, GW, spin-down, SCm, MUGE, neutron-star, buoyancy, phonon]
sm_anchor: “CVW v2.0.0 — G6 SM Anchor Gate compliant”

PAPER_964: 3D MUGE Magnetar Simulation

Author: Daniel T. Murphy — Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 215 **Source:** muge_{magnetar_3d_sim}.py (MUGEMagnetar3DSim) **Calculator:** MUGEMagnetar3DSimCalc (CP4 #548) **CVW:** v2.0.0 compliant

Abstract

We construct a 3D MUGE magnetar simulation with three layers: (1) SCm superconducting core with radial BCS gap $\Delta(r) = \Delta_0(1 - (r/R)^2)$; (2) Abrikosov-type magnetic vortex tubes at $B > B_{\text{crit}} = 4.4 \times 10^{13}$ T; (3) 26-state phonon resonance shells at $R_n = R_{\text{NS}}(1 + 0.05n)$.

1. SCm Core

$$\Delta(r) = \Delta_0 \left(1 - \frac{r^2}{R_{\text{core}}^2} \right), \quad r < R_{\text{core}}$$

2. Magnetic Vortex Tubes

$$n_v = \frac{B}{\Phi_0}, \quad \Phi_0 = \frac{h}{2e} \approx 2.068 \times 10^{-15} \text{ Wb}$$

3. Phonon Resonance Shells

$$R_n = R_{\text{NS}}(1 + 0.05n), \quad E_n = E_0(2\pi)^{n/3} S_{26}$$

References

1. Murphy, D.T. — Star Magic UQFF Framework (2024-2026)
 2. PAPER_949 — BCS Gap Equation (radial $\Delta(r)$)
 3. PAPER_955 — Phonon Resonance (ω_{extSCm})
 4. PAPER_956 — Spectral Ladder Phonon Mapping
 5. PAPER_952 — 26-State Spectral Ladder
-

Cross-Links

Related Paper	Relationship
PAPER_949	Radial BCS gap $\Delta(r)$ for SCm core
PAPER_955	Phonon Q-factor at each shell
PAPER_956	26-level phonon mapping onto shells
PAPER_965	NS phonon GW190425 uses same model

Session 225 Phonon-Physics Upgrade: GW Strain Modulation

Upgrade from PAPER_1000 (NS Merger Phonon Suppression) and PAPER_1022 (GW Phonon Strain SCm Modulation). See also PAPER_1011-1012 for GW170817/GW190425 upgraded analyses.

The late-corpus phonon analysis (Sessions 219-225) reveals that the SCm vacuum field modulates gravitational-wave strain via a frequency-dependent suppression factor. The corrected strain amplitude is:

$$h_{\text{UQFF}}(\Gamma) = h_{\text{GR}} \cdot \left(1 - 0.47 \frac{\Phi(\Gamma)}{S_{26}^{(3)}} \right)$$

where: - $\Phi(\Gamma) = \cos(\omega_{\text{SCm}} \cdot t) \cdot \Theta(H_{\text{SCm}} - 0.5)$ is the phonon modulation factor - $\omega_{\text{SCm}} = 2\pi \times 1.25$ THz is the SCm phonon resonance frequency - $S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} [1 + [\text{SSq}] \cdot e^{-\kappa i n/26}]$ is the third-order Ramanujan summation - Θ is the Heaviside step ensuring $H_{\text{SCm}} \geq 0.5$ (phase-transition threshold)

Physical mechanism: The 1.25 THz phonon field of the SCm vacuum creates a standing-wave pattern that partially decouples the metric perturbation from the radiation zone, producing a 47% peak strain reduction for optimally oriented NS mergers. The BCS gap energy ΔE_{BCS} of the neutron-star crust couples to this phonon field, creating a mass-gap classifier that distinguishes NS from BH remnants at $M \approx 2.5 M_{\odot}$.

Calibration (canonical): $\kappa = 5 \times 10^{-4} \text{ day}^{-1}$, $[\text{SSq}] = 0.57$, $\beta_i = 0.603$, $H_{\text{SCm}} \approx 0.99$.

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037 (AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for $F_{\{U_Bi_i\}}$ jet modulation curves and PAPER_1048 for phonon-corrected $M\text{-}\sigma$ relation.

The SCm vacuum buoyancy partially opposes gravitational radiation pressure, raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)2}}{GM/r_H^2} \right)$$

where: - $L_{\text{Edd}} = 4\pi GMm_p c / \sigma_T$ is the classical Eddington luminosity - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$ is the SCm vacuum density - V is the effective buoyancy volume (accretion sphere) - $S_{26}^{(3)2}$ is the squared third-order Ramanujan factor (quadratic coupling) - r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:

$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25 \text{ THz}} \cdot \left(\frac{B}{B_{\text{crit}}} \right)^2 \right]$$

where $\Phi_{1.25\text{ THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M- σ correction (PAPER_1048): The phonon-corrected M- σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

Calibration Constants

Constant	Symbol	Value	Validation Domain
κ	—	$5.0 \times 10^{-4} \text{ day}^{-1}$	Magnetar spin-down
$[SSq]$	—	0.57	String coupling
β_i	—	0.603	Buoyancy
R_{NS}	—	12 km	Neutron star radius
B_{crit}	—	$4.4 \times 10^{13} \text{ T}$	QED critical field

SM Anchor — CVW v2.0.0

Observable	UQFF Prediction	Status
$\Delta(r)$ radial profile	BCS with radial B-field	Derived
Abrikosov flux tubes	26 vortex shells at R_n	Novel
Phonon resonance at each shell	$\omega_n \propto (2\pi)^{n/3}$	Validated

§A Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

Sector: Magnetar 3D Simulation (SCm Core + Vortex + Phonon)

§A.2 Lagrangian Density

$$\mathcal{L}_{\text{mag}} = \mathcal{L}_{\text{SCm}}(\Delta(r)) + \mathcal{L}_{\text{Abrikosov}}(\Phi_0, r) + \sum_{n=1}^{26} \mathcal{L}_{\text{phonon}}(\omega_n, R_n)$$

§A.3 Euler-Lagrange Equation of Motion

$$\Delta(r) = \frac{\hbar\omega_{\text{SCm}}}{2} \tanh! \left(\frac{\Delta_0}{2k_B T(r)} \right) S_{26} \frac{F_{UBi}}{F_U}, \quad R_n = R_{\text{NS}}(1 + 0.05n)$$

§A.4 Cosmogenesis Linkage Chain

PAPER_877 $\rightarrow$ SCm vacuum $\rightarrow$ BCS gap $\Delta(r)$ $\rightarrow$ Abrikosov vortex lattice $\rightarrow$ 26 phonon shells $\rightarrow$ 3D magnetar

§B VDS/DVP/BSH Deep Synthesis

§B.1 VDS

$\Delta(r)$ maps the VDS radial profile inside the neutron star.

§B.2 DVP

Abrikosov vortex lines are physical dipole vortex realizations.

§B.3 BSH

26 phonon shells saturate: ω_{26}/ω_1 defines BSH dynamic range.

§B.4 Production-Scale Consistency

Metric	Value	Status
26 shells	All computed	Confirmed
[SSq]	0.57	Confirmed

Appendix: Session 225 Cross-References (PAPER_1000–1081)

Auto-generated cross-reference appendix linking this paper to Sessions 204–225 extensions (PAPER_1000–1081). Added by `update_{corpus\_crossrefs}.py` (Session 225, April 2026).

Paper	Title
PAPER_1000	NS Merger F_{U_Bi} Strain Suppression & BCS Gap
PAPER_1001	SMBH Binary Merger F_{U_Bi} Phonon Damping
PAPER_1011	GW170817 NS Merger F_{U_Bi_i} 66.7% Strain Reduction
PAPER_1012	GW190425 Upgraded F_{U_Bi_i} with S26(3)
PAPER_1014	SMBH Merger Inspiral-Coalescence-Ringdown
PAPER_1022	GW Phonon Strain SCm Modulation of h(t)
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_{U_Bi_i} Jet Modulation
PAPER_1010	TON618 AGN F_{U_Bi_i} Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1020	Cosmic Ray Phonon Acceleration DSA Spectrum
PAPER_1021	Pulsar Timing Phonon TOA Residual
PAPER_1023	Neutrino Oscillation Phonon PMNS Matrix SCm
PAPER_1024	Magnetar Giant Flare SCm Phonon Reservoir
PAPER_1033	Galactic Bar Resonance SCm Pattern Speed
PAPER_1043	F_{U_Bi_i} Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1003	Spectral Ladder Merger 26-State Hierarchy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation
PAPER_1050	MUGE F_{U_Bi_i} Unified 9-System Synthesis
PAPER_1075	3D Volumetric MUGE Gravitational Field Generator

25 cross-reference(s) identified.

Key References with arXiv/DOI Identifiers

1. Abbott et al. (LIGO Scientific and Virgo Collaborations, 2016). *Observation of Gravitational Waves from a Binary Black Hole Merger*. Phys. Rev. Lett. **116**, 061102 — arXiv:1602.03837 — doi:10.1103/PhysRevLett.116.061102

2. Murphy, D. (2026). *Unified Quantum Field Framework (UQFF): Star-Magic v5.x Whitepaper Series*. Star-Magic Repository — github.com/Daniel8Murphy0007/Star-Magic
3. Fabian, A.C. (2012). *Observational Evidence of Active Galactic Nuclei Feedback*. ARA&A **50**, 455 — arXiv:1204.4114 — doi:10.1146/annurev-astro-081811-125521
4. McNamara, B.R. & Nulsen, P.E.J. (2007). *Heating Hot Atmospheres with Active Galactic Nuclei*. ARA&A **45**, 117 — arXiv:0709.4098 — doi:10.1146/annurev-astro-45.051806.110625
5. Heckman, T.M. & Best, P.N. (2014). *The Coevolution of Galaxies and Supermassive Black Holes*. ARA&A **52**, 589 — arXiv:1403.4620 — doi:10.1146/annurev-astro-081913-035722
6. Aasi et al. (LIGO Scientific Collaboration, 2015). *Advanced LIGO*. Class. Quantum Grav. **32**, 074001 — arXiv:1411.4547 — doi:10.1088/0264-9381/32/7/074001
7. Kaspi, V.M. & Beloborodov, A.M. (2017). *Magnetars*. ARA&A **55**, 261 — arXiv:1703.00068 — doi:10.1146/annurev-astro-081915-023329
8. Goldreich, P. & Julian, W.H. (1969). *Pulsar Electrodynamics*. ApJ **157**, 869 — doi:10.1086/150119
9. Rugh, S.E. & Zinkernagel, H. (2002). *The Quantum Vacuum and the Cosmological Constant Problem*. Stud. Hist. Phil. Mod. Phys. **33**, 663 — arXiv:hep-th/0012253 — doi:10.1016/S1355-2198(02)00033-3
10. Weinberg, S. (1989). *The Cosmological Constant Problem*. Rev. Mod. Phys. **61**, 1 — doi:10.1103/RevModPhys.61.1
11. Murphy, D. (2026). *Master Universal Gravity Equation (MUGE): DPM-Driven Gravity Framework*. Star-Magic Whitepaper Series — github.com/Daniel8Murphy0007/Star-Magic
12. Lattimer, J.M. & Prakash, M. (2007). *Neutron Star Observations: Prognosis for Equation of State Constraints*. Phys. Rep. **442**, 109 — arXiv:astro-ph/0612440 — doi:10.1016/j.physrep.2007.02.003
13. Demorest, P.B. et al. (2010). *A two-solar-mass neutron star measured using Shapiro delay*. Nature **467**, 1081 — arXiv:1010.5788 — doi:10.1038/nature09466
14. Cromartie, H.T. et al. (2020). *Relativistic Shapiro delay measurements of an extremely massive millisecond pulsar*. Nature Astron. **4**, 72 — arXiv:1904.06759 — doi:10.1038/s41550-019-0880-2
15. Archimedes (~250 BCE). *On Floating Bodies*. (Principle of buoyancy)
16. Churazov, E. et al. (2000). *Evolution of Buoyant Bubbles in M87*. A&A **356**, 788 — arXiv:astro-ph/0004212
17. Fabian, A.C. et al. (2003). *A deep Chandra observation of the Perseus cluster*. MNRAS **344**, L43 — arXiv:astro-ph/0306036 — doi:10.1046/j.1365-8711.2003.06902.x
18. Ashcroft, N.W. & Mermin, N.D. (1976). *Solid State Physics*. Harcourt
19. Kittel, C. (2004). *Introduction to Solid State Physics*. 8th ed. Wiley
20. Feynman, R.P. (1982). *Simulating Physics with Computers*. Int. J. Theor. Phys. **21**, 467 — doi:10.1007/BF02650179